Generalization in LMs

- Much of the criticism against Markov LMs (and about statistical methods to Computational Linguistics) has focused on the difficulty of such models in generalizing beyond the observed data
- Smoothing is a simple example for such a generalization
- But what about?

“Colorless green ideas sleep furiously”

vs.

“Furiously sleep ideas green colorless”

Chomsky (1957)

Pereira’s Solution

- Now assume that when a word appears, it is accompanied by a category, but that this category is never observed
- The identity of the next word is determined based on this category
  - So words share a category if they share a distribution of the next word
- Examples:
  - The words “the” and “a” may share a category, because they are both followed by nouns and verbs

the/a cat
the/a dog
the/a university

Formal grammar and information theory: together again? F. Pereira, Phil. Trans. R. Soc. Lond. A (2000)

Pereira's Solution

- In probabilistic terms we will say that each word w_i is accompanied by a latent category c_i such that

$$Pr(w_i|w_{i-1}) = \sum_{c \in CATS} Pr(w_i, c|w_{i-1}) = \sum_{c \in CATS} Pr(c|w_{i-1}) \cdot Pr(w_i|c, w_{i-1}) =$$

$$\sum_{c \in CATS} Pr(c|w_{i-1}) \cdot Pr(w_i|c)$$

  - The last transition assumes that w_{i-1} and w_i are conditionally independent given c
- The model is now free to assign words to categories, such that knowing the category of a word is the best predictor of the following word

Pereira’s Solution

- Using 16 categories and newspaper corpora to train the model (using EM), Pereira found that:

$$\frac{\Pr(Colorless\ green\ ideas\ sleep\ furiously)}{\Pr(Furiously\ sleep\ ideas\ green\ colorless)} \approx 200000$$

- Today’s LMs use similar technology to generalize to novel word sequences, by creating a multi-dimensional representation of each word